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Instructor:	Samuel Chukwuemeka		
Project:	Akron Ohio Water Bill – Piecewise Functions		
Company:	City of Akron – Water and Sewer Rates (https://www.akronohio.gov/cms/site/dced0363af40e9b7/index.html)		
Objectives:	1.) Calculate the spring water bill of Akron city's residents depending on their water usage, using first the arithmetic method and then piecewise functions. 2.) Write a piecewise function for different residential water rates. 3.) Calculate the power bill using random water usage values and plug them into a piecewise function.		
Information:	Current Akron Water Rates		
	Block 1	0 – 30 hcf	\$2.73 per hcf
	Block 2	31 – 5,000 hcf	\$2.45 per hcf
	Block 3	5,001 and over hcf	\$1.29 per hcf
	Billing Charge Fee Per Month: \$4.00		
Numbers to Test:	1.) 0 hcf 2.) 20 hcf 3.) 100 hcf 4.) 3,000 hcf 5.) 6,000 hcf 6.) 7,000 hcf		
Arithmetic Method:	The water monthly bill fee is not a piece. The application has three pieces Let w = Water used Let c = Cost per hcf of water used in \$ 1. 0 hcf falls in the first piece. Monthly Bill Fee = \$4.00 Rate for 0 hcf: \$2.73 per hcf $0 * 2.73 = 0$ $0.00 + 4.00 = 4.00$ The cost for 0 hcf is \$4.00. 2. 20 hcf falls in the first piece. Monthly Bill Fee = \$4.00 Rate for 20 hcf: \$2.73 per hcf $20 * 2.73 = 54.60$ $54.60 + 4.00 = 58.60$ The cost for 20 hcf is \$58.60		

3. 100 hcf falls in the second piece.

Monthly Bill Fee = \$4.00

Before we are able to use the second piece, we need to go through the previous piece first.

The cost for 30 hcf * 2.73 = \$81.90

$100\text{hcf} - 30\text{hcf} = 70\text{hcf}$

Now that we have the remaining, we need to find the cost for 70 hcf which takes us to the second piece.

The second piece for 70 hcf

Rate for 70 hcf: \$2.45 per hcf

$70 * 2.45 = 171.50$

$171.50 + 81.90 + 4.00 = 257.40$

Cost for 100 hcf = \$257.40

4. 3,000 hcf falls in the second piece.

Monthly Bill Fee = \$4.00

Before we are able to use the second piece, we need to go through the previous piece first.

The rate for 30 hcf: \$2.73 per hcf

$2.73 * 30 = \$81.90$

$3,000 - 30 = 2,970$

Now that we have the first 30 hcf we need to find the cost for the remaining 2,970 hcf

That takes us to the second piece.

Second piece for 2,970 hcf

Rate for 2,970 hcf: \$2.45 per hcf

$2.45 * 2,970 = 7,276.50$

$7,276.50 + 81.90 + 4.00 = 7,362.40$

Cost for 3,000 hcf = \$7,362.40

5. 6,000 hcf falls in the third piece

Monthly Bill Fee = \$4.00

Before we are able to use the third piece, we need to go through the two previous pieces.

First piece for 30 hcf

Rate for 30 hcf: \$2.73 per hcf

$$2.73 * 30 = 81.9$$

$$5,000 - 30 = 4,970 \text{ hcf}$$

Second piece for 4,970 hcf

Rate for 4,970 hcf: \$2.45 per hcf

$$2.45 * 4970 = 12,176.50$$

$$6000 - 5000 = 1,000$$

Now that we have the first and second pieces, we need to find the cost for the remaining 1,000 hcf

Third piece for 1,000 hcf

Rate for 1,000 hcf: \$1.29 per hcf

$$1,000 * 1.29 = 1,290$$

$$1,290 + 81.90 + 12,176.50 + 4.00 = 13,552.40$$

$$\text{Cost for 6,000 hcf} = \$13,552.40$$

6. 7,000 hcf falls in the third piece

Monthly Bill Fee = \$4.00

Before we are able to use the third piece, we need to go through the two previous pieces.

First piece for 30 hcf

Rate for 30 hcf: \$2.73 per hcf

$$2.73 * 30 = 81.90$$

$$5,000 - 30 = 4,970 \text{ hcf}$$

Second piece for 4,970 hcf

Cost for 4,970 hcf: \$2.45 per hcf

$$2.45 * 4970 = 12,176.50$$

$$7000 - 5000 = 2,000$$

Now that we have the first and second pieces, we need to find the cost for the remaining 2,000 hcf

Third piece for 2,000 hcf

Rate for 2,000 hcf: \$1.29 per hcf

$$2,000 * 1.29 = 2,580.00$$

$$2,580.00 + 81.90 + 12,176.50 + 4.00 = \$14,842.40$$

$$\text{Cost for 7,000 hcf} = \$14,842.40$$

Piecewise
Function:

Let w = Water consumed (in hcf)
Let c = Cost per hcf or water consumed (in \$)

For the first piece

Basic service fee = \$4

Cost for w hcf: 2.73 per hcf = $w \cdot 2.73 = 2.73w$

$$c(w) = 2.73w + 4$$

For the second piece

Basic service fee = \$4

We have to finish with the first piece first

Cost for 30 hcf = 2.73 per hcf = $30 \cdot 2.73 = 81.9$

$$4 + 2.73(30) + 2.45(w - 30)$$

$$4 + 81.9 + 2.45w - 73.5$$

$$2.45w + 12.4$$

$$c(w) = 2.45w + 12.4$$

For the third piece

Basic service fee = \$4

We have to finish with the first piece first

Cost for 30 hcf = 2.73 per hcf = $30 \cdot 2.73 = 81.9$

Then we have to finish with the 2nd piece next

$$4 + 2.73(30) + 2.45(5000 - 30) + 1.29(w - 5000)$$

$$4 + 81.9 + 12,262.4 + 1.29w - 6450$$

$$5,812.4 + 1.29w$$

$$c(w) = 1.29w + 5,812.4$$

$$c(w) = \begin{cases} 2.73w + 4; & 0 \leq w \leq 30 \\ 2.45w + 12.4; & 30 < w \leq 5,000 \\ 1.29w + 5,812.4; & w > 5,000 \end{cases}$$

Numbers
Tested
using
algebraic
method:

0 hcf: First piece: $2.73(0) + 4.00 =$
 $0.00 + 4.00 =$
\$4.00 for 0hcf of water used.

20 hcf: First piece: $2.73(20) + 4 =$
 $54.60 + 4.00 =$
\$58.60 for 30hcf of water used.

100 hcf: Second piece: $2.45(100) + 12.4 =$
 $245.00 + 12.40 =$
\$257.40 for 100hcf of water used.

3,000hcf: Second piece: $2.45(3000) + 12.4 =$

$7,350.00 + 12.40 =$
 $\$7,362.40$ for 3,000hcf of water used.

6,000hcf: Third piece: $1.29(6000) + 5812.4 =$
 $7740.00 + 5812.40 =$
 $\$13,552.40$ for 6,000hcf of water used

7,000hcf: Third piece: $1.29(7000) + 5812.4 =$
 $9030.00 + 5812.40 =$
 $\$14,842.40$ for 7,000hcf of water used

Citations:

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Chuckwuemeka, S.D (2018). *Piecewise functions*. Piecewise Functions. Retrieved February 22, 2023, from [https://piecewise- functions.appspot.com/#powerBill](https://piecewise-functions.appspot.com/#powerBill)

Customer service. Water and Sewer Rates: City of Akron. (2016). Retrieved February 22, 2023, from <https://www.akronohio.gov/cms/site/dced0363af40e9b7/index.html>